

Data Mining Complete Exam Notes

CT 72502 (Elective I) · BEI / BEX IV-I · TU, IOE · tiers over 17 papers, 2070–2081 BS

7. Advanced Applications

3 hrs · in 16/17 papers · 3 topics · 1 numerical (PageRank) → Ch 7 companion.

Weight of the chapter:

- Web mining **TOP 11/17** → **3 hrs of syllabus, asked in 11 papers. Best marks-per-hour after Ch 6.**
 - Time-series data mining **HOT 6/17** · Mining object & multimedia **PIN 3/17**
 - Mostly **short notes**, so breadth beats depth here.
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7.1 Web Mining

What is Web Mining? Briefly explain the structure / taxonomy of Web Mining. 3+5 6 8

TOP 11/17 81 Ba, 80 Bh, 79 Bh, 78 Bh, 76 Ch, 76 Ash, 75 Ch, 74 Ch, 74 Ash, 73 Shr, 70 Ch

Definition

- **Web mining** = applying data mining to discover patterns from **web data**: the content, the link structure and the usage logs.

Why the Web is hard (good opening lines)

- Huge, growing, and **widely distributed**.
- **Semi-structured / unstructured** (HTML, not tables).
- **Highly dynamic**; heterogeneous; multilingual.
- Only a **tiny fraction** is relevant to any one user.
- Broad topics return hundreds of thousands of documents → **search returns too much, and misses too much**.

The taxonomy (3 branches — this IS the answer)

- 1. Web Content Mining**: mine the *content* of pages (text, images, video).
 - *Web page content mining*: extract facts from within pages.
 - *Search result mining*: summarise / cluster / categorise results.
- 2. Web Structure Mining**: mine the *hyperlink graph*.
 - Links carry authority: a link is a **vote**.
 - Algorithms: **PageRank**, **HITS**, **CLEVER**.
 - Finds authoritative pages, hubs, and web communities.
- 3. Web Usage Mining**: mine the *server logs* / click-streams.
 - *General access pattern tracking*: how the site is used overall.
 - *Customised usage tracking*: per-user personalisation.
 - Uses: site restructuring, recommendation, targeted ads, caching, fraud detection.

PageRank (asked as a short note in 2 papers + 1 numerical)

- Ranks a page by **who links to it**, not by what it says.
- **Two ideas**:
 1. A link from page T to page A is a **vote** for A .
 2. Votes are **weighted by the voter's own rank**, and **split** across all its outgoing links.
- **Recursive**: a page is important if important pages point to it.

$$PR(A) = (1 - d) + d \sum_i \frac{PR(T_i)}{C(T_i)}$$

- T_i = pages linking to A ; $C(T_i)$ = number of **outbound links** from T_i .
- d = **damping factor**, usually **0.85**.
- **Random surfer model**: with probability d the surfer follows a link, with probability $(1 - d)$ they jump to a random page.
- **Why damping is needed**: without it, a group of pages linking only to each other would hoard all the rank (a "rank sink"), and pages with no outlinks would leak it.
- Computed **iteratively** until the values settle; ranks converge quickly.

HITS (for contrast)

- Two scores per page: **authority** (pointed to by good hubs) and **hub** (points to good authorities).
- **Query-dependent** and computed at query time; PageRank is query-independent and precomputed.

Text mining 81 Ba 4

- **Key steps:**
 1. **Text preprocessing:** tokenise, remove **stop words**, **stemming** (reduce to root form), case folding.
 2. **Feature/attribute generation:** build the **term vector** (bag of words).
 3. **Feature selection:** drop uninformative terms; weight by **TF-IDF**.
 4. **Mining:** classification, clustering, association, topic modelling.
 5. **Analysis / interpretation** of the results.
- Similarity between documents → **cosine similarity** → Ch 2, §2.4.
- **Information retrieval** measures: precision and recall → Ch 3, §3.7.

Challenges of web mining 75 Ch 5

- Scale and growth rate; dynamic content; the deep/hidden web.
- Noise: ads, navigation bars, boilerplate.
- **Spam:** link farms deliberately gaming the ranking.
- Privacy concerns in usage mining; no schema; duplicates and mirrors.

Mining object and multimedia data 6 3 4

PIN 3/17 75 Ash, 74 Ch, 75 Ch

- **Multimedia data** = image, audio, video, text-plus-media in digital form.
- **Feature extraction first** — you never mine the raw pixels:
 - Image: colour histogram, texture, shape, edges.
 - Audio: pitch, frequency, MFCC. Video: motion vectors, key frames.
- **Tasks:** similarity search (content-based retrieval), classification, clustering, association among media features, video-shot detection.
- **Challenges:** very high dimensionality, huge volume, and the **semantic gap** — low-level features do not map cleanly to human meaning.

Visual data mining 75 Ch short note 4

- Uses **visualisation** to help the user discover knowledge, combining human perception with machine computation.
 - **Categories:** data visualisation, mining-*result* visualisation, mining-*process* visualisation, and interactive visual mining.
 - Techniques: scatter plots, box plots, parallel coordinates, pixel-oriented, dendrograms.
 - **Why it works:** humans spot patterns and outliers in a picture far faster than in a table.
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7.2 Time-Series Data Mining

Explain time series data mining with an example. 6 5 4 3

HOT 6/17 80 Ba, 75 Ch, 74 Ch, 72 Ch, 72 Ka, 71 Shr Mostly short notes.

Definition

- A **time series** is a sequence of values recorded at **successive, usually equal, time intervals**.
- **Time-series data mining** = finding patterns, trends and regularities in such data.
- Eg stock prices, daily temperature, sales per month, sensor readings, ECG, network traffic, power consumption.
- **What makes it different:** the **order matters** and consecutive values are **correlated**, so the usual “independent tuples” assumption breaks.

The 4 components of a time series (memorise: T-C-S-I)

- **Trend (T):** the long-term overall direction. Eg steady sales growth.
- **Cyclic (C):** long, irregular swings, over years. Eg business cycles.
- **Seasonal (S):** regular, calendar-fixed variation. Eg umbrella sales in monsoon.
- **Irregular / random (I):** residual noise left after the other three.
- Models: multiplicative $Y = T \times C \times S \times I$, or additive $Y = T + C + S + I$.

Tasks

- **Trend analysis + forecasting** (moving average, exponential smoothing, ARIMA).
- **Similarity search:** find series that behave alike.
 - *Whole-sequence* matching vs *subsequence* matching.
 - Needs **normalisation** (offset and amplitude) first, else scale dominates.
 - **Dynamic Time Warping (DTW)** aligns series that are out of phase or run at different speeds — Euclidean distance cannot. verified/added
- **Motif discovery:** repeated subsequences. **Anomaly detection:** unusual subsequences → Ch 6.
- **Segmentation and classification** of series.

Smoothing: moving average

- Replace each value by the mean of a window around it → removes the irregular component and exposes the trend.
- A larger window smooths more but lags more.

Example to quote

- Daily closing prices of a share over 5 years: the **trend** is the multi-year rise, the **seasonal** part is the recurring festival-quarter bump, the **cyclic** part is the multi-year market swing, and a one-day crash is **irregular**. Mining finds the trend for forecasting and the crash as an anomaly.

7.3 PYQ Mapping

N = solved in the Ch 7 Numerical companion.

7.1 Web mining

1. **TOP 4/17** What is web mining + its structure / categories / taxonomy. 3+5 80 Bh 6 74 Ash 8 76 Ash 5 75 Ch
2. **PIN 1/17** Key steps in **text mining** + how to find page rank. 4+4 81 Ba
3. **N PageRank** with damping factor 0.85, 5 iterations, initial rank 0.25. 8 76 Ch
4. **PIN 2/17** Short notes: Page Rank algorithm 5 79 Bh 4 78 Bh
5. **HOT 3/17** Short note: Web mining 4 73 Shr 3 74 Ch 70 Ch

7.2 Mining object and multimedia

6. **PIN 1/17** Explain Web mining and Multimedia mining. 6 75 Ash
7. **PIN 2/17** Short notes: Multimedia mining 3 74 Ch; Visual Data Mining 4 75 Ch

7.3 Time-series

8. **PIN 1/17** Explain time series data mining in brief. 6 72 Ka
9. **PIN 1/17** Challenges of web mining + time series data mining w/ example. 5 75 Ch
10. **HOT 4/17** Short note: Time Series Data Mining. 5 80 Ba 3 74 Ch 4 72 Ch 71 Shr

Coverage check

- All Ch 7 PYQs covered. 1 numerical 76 Ch → companion.
- **This chapter is 3 hrs of syllabus but appears in 16/17 papers**, almost entirely as **4–6 mark short notes**. Learn the **web mining taxonomy**, the **PageRank formula** and the **4 time-series components** and you can answer nearly every one of them.
- **Gaps filled by research** verified/added : the **PageRank formula and the random-surfer justification for damping** (your slides name PageRank but never give the equation); **DTW**; the **T-C-S-I decomposition**; the **semantic gap** in multimedia mining.

Sources used for this chapter

- **Han & Kamber 3rd ed** ch 13 (trends and applications) and the mining-complex-data material. **Primary authority.**
- Chapter9_0.ppt (web mining taxonomy), Chapter9_1.ppt (multimedia), Chapter9_2.ppt (text and web), a1VisualMine.ppt (visual data mining), data_mining_/Chapter 6 _ 7.pdf, **DBK Lectures 18–19**.

Abbreviations used in these notes

Only these are ever shortened. Anything not on this list is written out in full.

Short	Means	Short	Means
DM	data mining	DW	data warehouse
DB	database	DBMS	database management system
KDD	knowledge discovery in databases	DSS	decision support system
OLTP	on-line transaction processing	OLAP	on-line analytical processing
ROLAP	relational OLAP	MOLAP	multidimensional OLAP
HOLAP	hybrid OLAP	ETL	extract, transform, load
BI	business intelligence	ER	entity–relationship
attr(s)	attribute(s)	dim(s)	dimension(s)
multidim	multidimensional	txn(s)	transaction(s)
info	information	ops	operations
repo	repository	freq	frequent / frequency
config	configuration	no.	number
PCA	principal component analysis	SVD	singular value decomposition
SMC	simple matching coefficient	KNN	<i>k</i> -nearest neighbour
ANN	artificial neural network	POS	point of sale
FK	foreign key	UI	user interface
GIGO	garbage in, garbage out	w/, w/o	with, without
rm	remove	vs	versus
Eg	example / for example	+Fig	draw a figure
→	leads to / therefore	+	and

Exam-paper month codes (used in the year tags): Ba = Baishakh · Jth = Jestha · Asa = Ashar · Shr = Shrawan · Bh = Bhadra · Ash = Ashwin · Ka = Kartik · Mng = Mangsir · Po = Poush · Ma = Magh · Ch = Chaitra.

Tier chips: **TOP /17** = topic asked in 7 or more of the 17 papers · **HOT /17** = 4–6 papers · **PIN /17** = 1–3 papers but worth ≥8 marks. **Mark chips** such as **2+5** are the marks that question actually carried. verified/added marks anything researched and added that was not in the source notes.